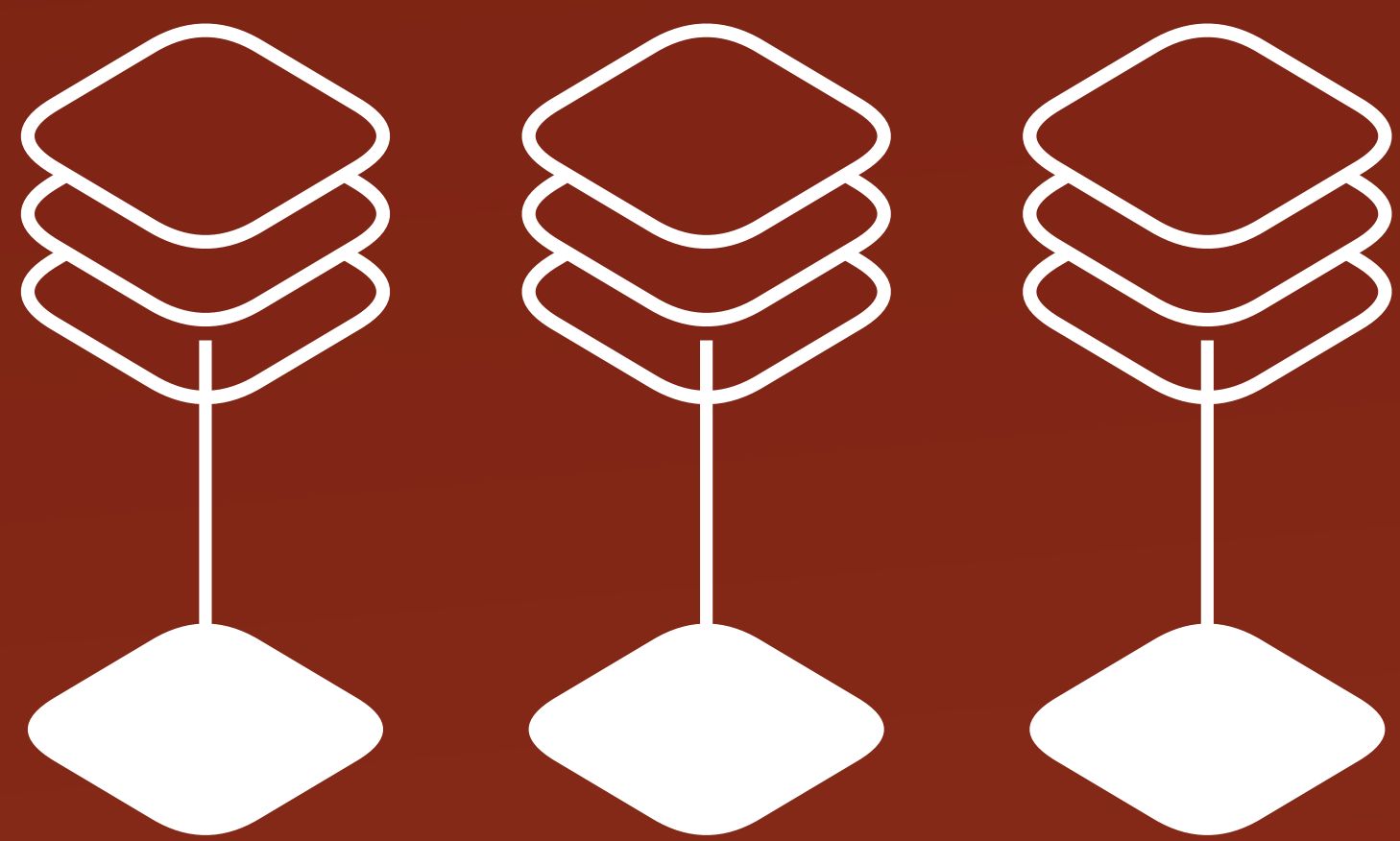
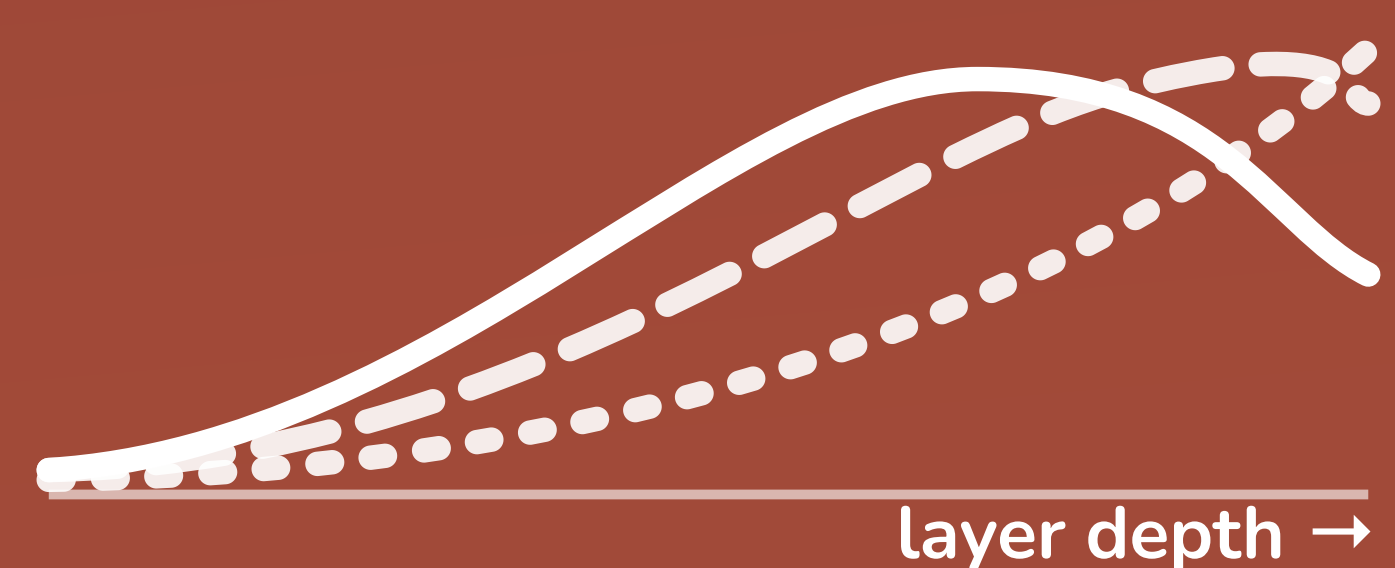




Which layer of a frozen vision foundation model should you probe?

None alone.

For dense prediction, try **AMDF**: it fuses **all** layers, for **every** patch.



On DINOv3-L, NYUv2 probes **peak at different depths**: normals L18, depth L21, segmentation L24.

1.48

Best mean rank over 5 benchmarks (4 dense + camera pose) and 7 frozen backbones.

$\approx 1/4$

of the **trainable parameters** of concatenating all layers (1.6M vs 6.8M, DINOv3-L, normals).

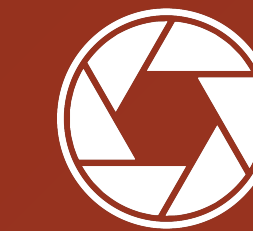
Frozen backbone · no layer search · one probe, any head

# AMDF: Holistic Probing of Vision Foundation Models with Adaptive Multi-Layer Dense Fusion

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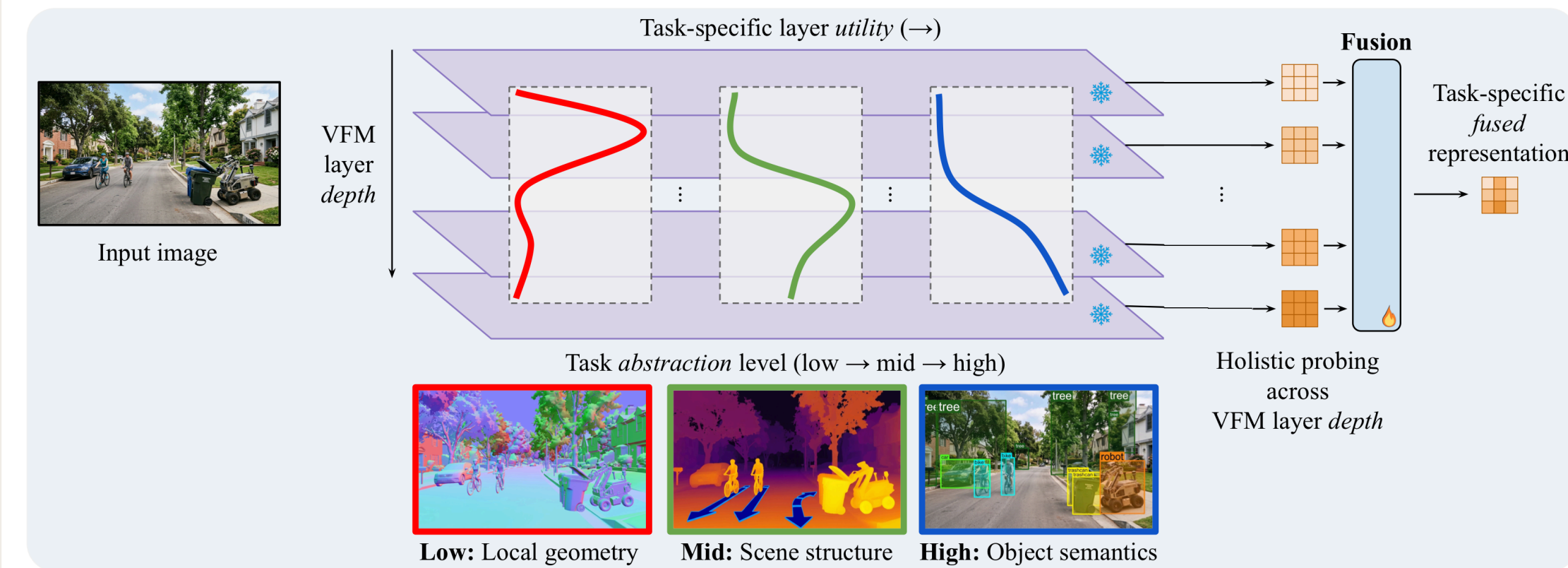
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Copernicus Science Centre  
Warsaw · October 8–11, 2026

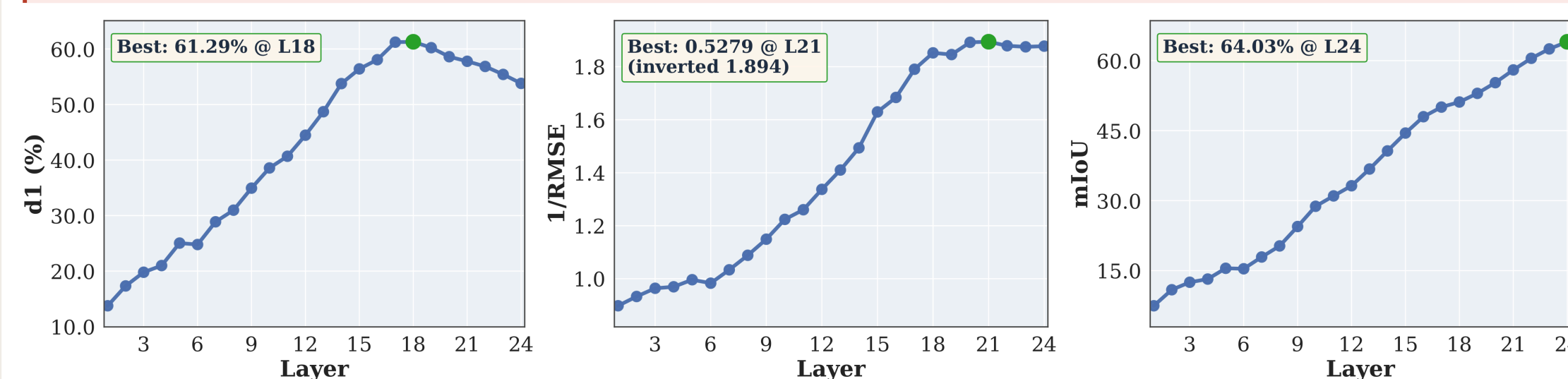
## Different tasks live at different depths



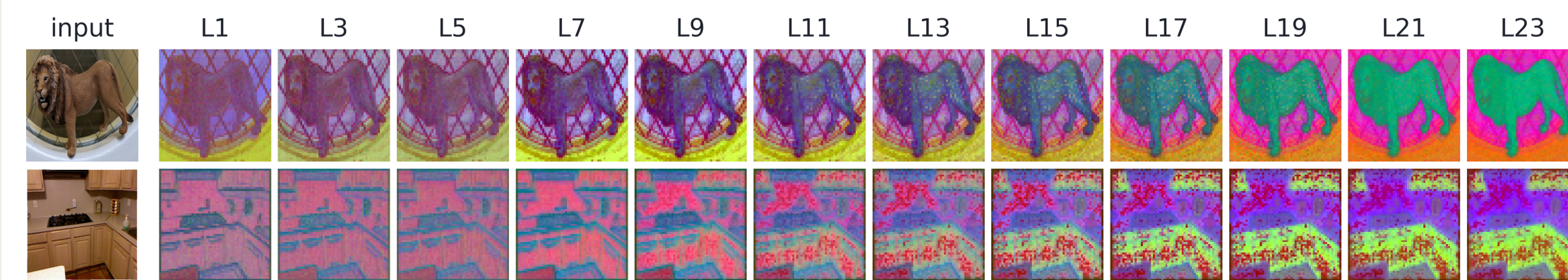
**Fig. 1:** Geometry peaks in middle layers, semantics near the end. AMDF reads every frozen layer and fuses them per patch.

## Problem #1: the last layer is not the best

**X** Probing only the final layer misses what a task needs.



**Fig. 2:** One probe per layer of a frozen DINOv3-L on NYUv2: the best layer moves deeper as the task gets more semantic (L18 → L21 → L24).



**Fig. 3:** Layers see differently: frozen PESpatial-L patch features (one PCA per image, 770 px input); object-background separation appears only in deep layers.

## Problem #2: using all layers is costly

**X** Concatenating all layers works, but needs **6.8M trainable parameters** (vs **1.6M** for AMDF, DINOv3-L, incl. head) — and still treats every patch the same.

✓ **AMDF** learns, for every patch, how much of each layer to use.

## Setup

### 7 frozen backbones

DINOv3-S/B/L · MAE-L · SigLIP2-L · V-JEPA2.1-L · PESpatial-L

### 6 benchmarks

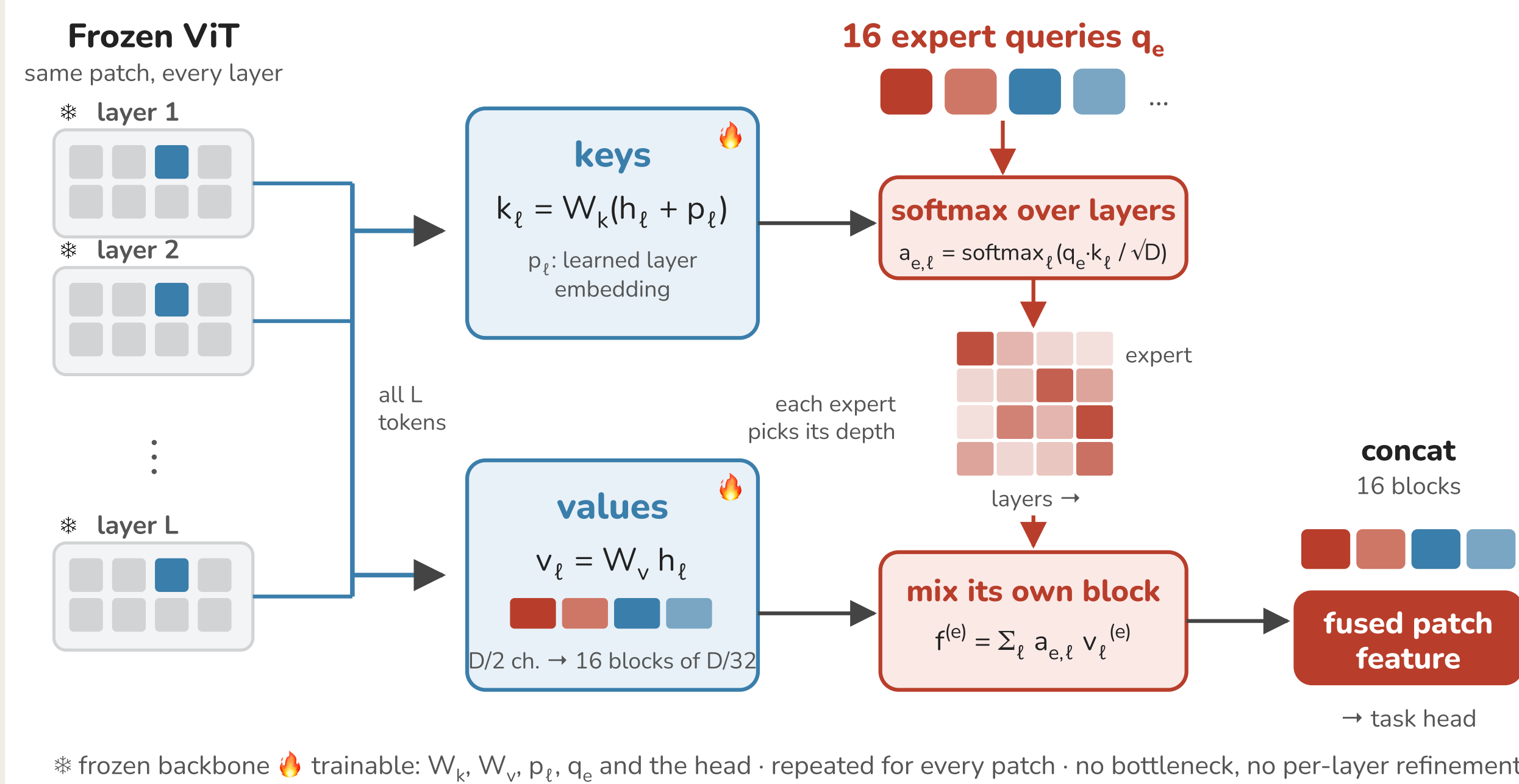
NYUv2 normals, depth, segmentation · Taskonomy · NAVI camera pose · CUB-200

### Fair comparison

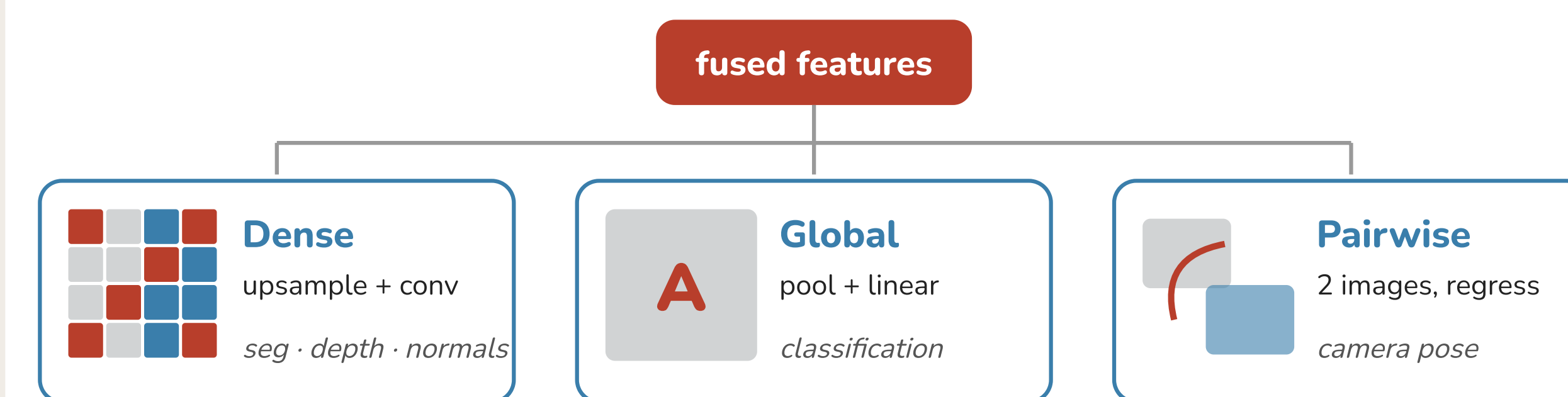
same recipe for every row of a table · mean of up to 3 seeds (a few cells have 1–2) · best-epoch validation metric

## Solution: AMDF — depth-aware fusion

For every patch, 16 learned expert queries each take a **softmax over all layers** and mix their own block of channels, so each expert picks its own depth.

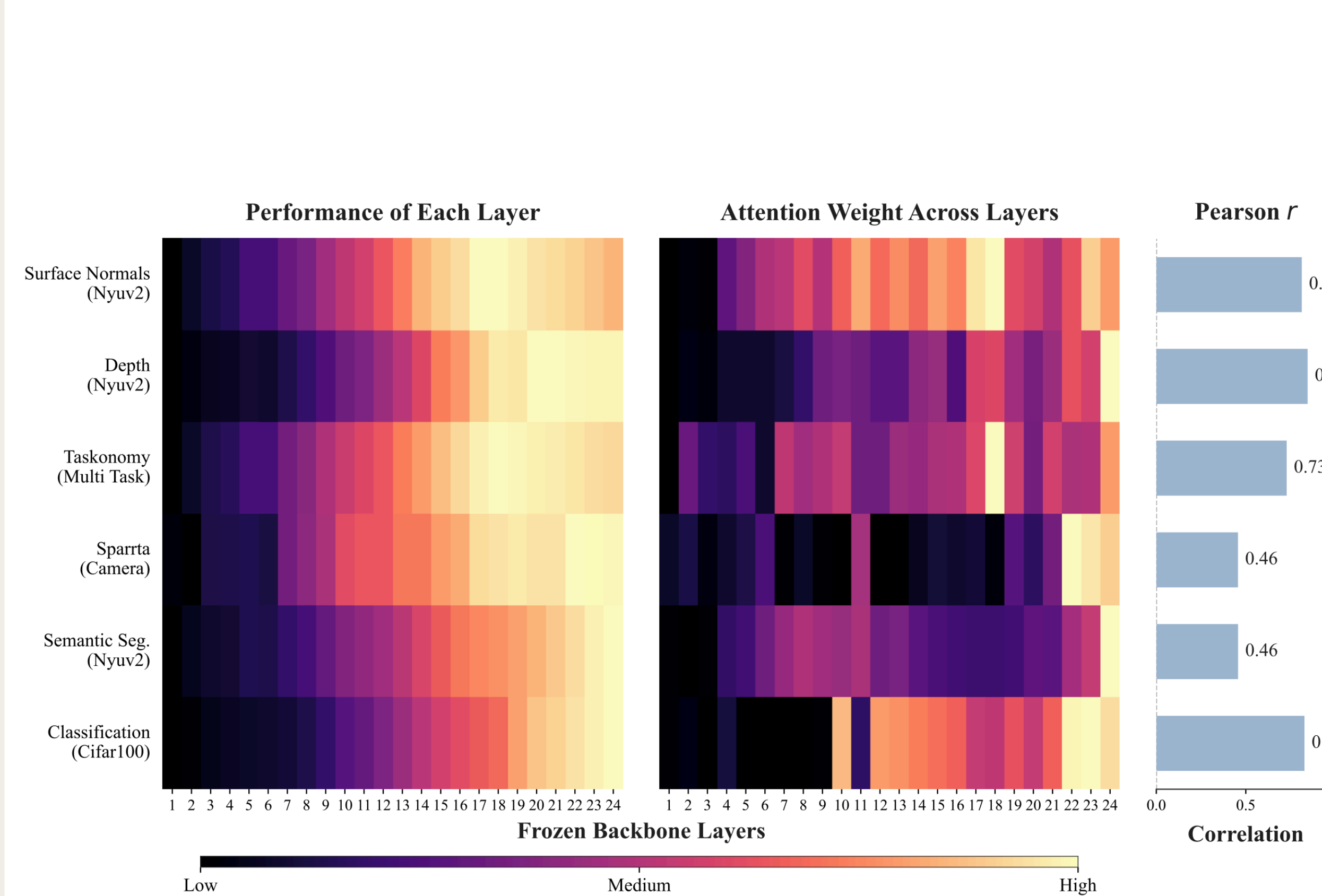


## One probe, three kinds of heads



**Fig. 4:** The backbone stays frozen and all ViT blocks are used; only the fusion probe and the task head are trained.

## Its attention follows layers utility



**Fig. 5:** Measured on an earlier AMDF variant (with per-layer refinement). AMDF's learned depth-attention (middle) tracks per-layer probe performance (left) on DINOv3-L/16. Alignment (Pearson  $r$ , right) is strongest for dense geometric tasks (depth  $r \approx 0.85$ , normals  $r \approx 0.81$ , Taskonomy  $r \approx 0.73$ ) and shifts toward deep layers for semantic tasks (classification  $r \approx 0.83$ ).

## Results

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benchmarks where AMDF ranks best

16

expert queries, each picking its own depth

7×6

backbones × benchmarks

Mean rank per benchmark (1 = best, ★ = best in table)

|                    | Normals       | Depth         | Segm.         | Taskonomy     | Pose          | CUB-200       |
|--------------------|---------------|---------------|---------------|---------------|---------------|---------------|
| <b>AMDF (ours)</b> | <b>1.00 ★</b> | 1.86          | <b>1.21 ★</b> | <b>1.29 ★</b> | 2.04          | 2.43          |
| ConcatLayers       | 2.00          | <b>1.14 ★</b> | 2.43          | 1.71          | <b>1.32 ★</b> | 4.57          |
| WeightedLayers     | 3.14          | 3.43          | 3.86          | 3.14          | 3.43          | 4.71          |
| LastLayer          | 4.57          | 4.00          | 3.21          | 4.43          | 4.25          | –             |
| AvgLayer           | 4.29          | 4.57          | 4.29          | 4.43          | 3.96          | 6.29          |
| EfficientProbe     | –             | –             | –             | –             | –             | <b>1.14 ★</b> |
| AttentiveFusionAP  | –             | –             | –             | –             | –             | 3.00          |
| AvgPool            | –             | –             | –             | –             | –             | 5.86          |

**Fig. 6:** Each cell is a method's rank averaged over the 7 backbones of that table (NAVI: rotation and translation). AMDF is best on normals, segmentation and Taskonomy; ConcatLayers on depth and pose; EfficientProbe on CUB-200. Full tables: handout.

## Accuracy vs. trainable parameters

|                    | Mean rank ↓ (4 dense + pose) | Trainable params (M) |
|--------------------|------------------------------|----------------------|
| <b>AMDF (ours)</b> | <b>1.48</b>                  | <b>1.79</b>          |
| ConcatLayers       | 1.72                         | 6.99                 |
| WeightedLayers     | 3.40                         | 0.70                 |
| LastLayer          | 4.09                         | 0.70                 |
| AvgLayer           | 4.31                         | 0.70                 |

**Fig. 7:** Mean of the per-table ranks over 5 benchmarks (NYUv2 normals, depth, segmentation, Taskonomy, NAVI pose); params (probe + head, ViT-L) averaged over the same tasks. CUB-200 excluded (different methods).

## Take-home

- Useful features are **spread across depth**, and the best depth depends on the task.
- AMDF has a better mean rank than LastLayer, AvgLayer and WeightedLayers in every table, and leads ConcatLayers on normals, segmentation and Taskonomy with **1/4 of its parameters**.
- On fine-grained classification, a dedicated last-layer probe (EfficientProbe) remains stronger.
- Study frozen VFMs as **full hierarchies**; no manual layer search needed.